Avi Wadhwa

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Education

Tata Institute of Fundamental Research (TIFR) — Bengaluru, IN-KA
International Centre for Theoretical Sciences (ICTS), Ph.D. in Physics — 2021–Current

- Thesis Topic: "Post-Newtonian expansion in anti-de Sitter spacetime and the dual conformal field theory."
- Thesis Advisor: Loganayagam R, ICTS-TIFR, Bengaluru.

Birla Institute of Technology and Science (BITS), Pilani — Sancoale, IN-GA
K.K. Birla Goa Campus, Integrated M.Sc. (Hons) in Physics — 2014–2018

- Thesis Topic: "Aspects of Superconformal Representation Theory in Four Dimensions."
- Thesis Advisor: Abhijit Gadde, TIFR, Mumbai.

Experience

Teaching Assistant - Summer School — Bengaluru, IN-KA
Summer School on Gravitational Wave Astronomy — 2026
Tutored for the course on "Effective Field Theory approach to Post-Newtonian Gravity" by Jan Steinhoff.

ICTS String Journal Club — Bengaluru, IN-KA
Gave lectures on conformal field theory and conformal bootstrap. — 2025

Teaching Assistant - Graduate Course — Bengaluru, IN-KA
Tutored for the "Introduction to General Relativity" graduate course at ICTS by Ashoke Sen. — 2023

Visiting Student — Bengaluru, IN-KA
Indian Institute of Science — 2020

Visiting Student — Bhubaneswar, IN-OR
Institute of Physics — 2019

Visiting Student (Undergraduate Thesis) — Mumbai, IN-MH
Tata Institute of Fundamental Research — 2018

Member - SEDS Celestia — Goa, IN-GA
Astronomy and astrophysics club of BITS Pilani - Goa campus — 2016

Publications and Preprints

[1] R. Loganayagam and **A. Wadhwa**, "Post-Newtonian theory in de Sitter and anti-de Sitter spacetimes", Oct. 2026. arXiv: 2610.20589 [hep-th].

[2] P. Chattopadhyay and **A. Wadhwa**, "Subleading Chern-Simons soft factors in perturbative de Sitter", May 2026. arXiv: 2605.05130 [hep-th].

[3] **A. Wadhwa**, "Soft theorems in Chern-Simons matter theories", *Phys. Rev. D*, vol. 113, no. 4, p. 045 017, 2026. arXiv: 2509.02542 [hep-th].

Programs, Schools and Conferences Attended

- Summer School on Gravitational Wave Astronomy, ICTS-TIFR, Bengaluru, India, 2026.
- Strings 2026, Shanghai, China. **Poster Presentation.**
- Pre-Strings school, SIMIS, Shanghai, China, 2026. **Poster Presentation.**
- School for Advanced Topics in Particle Physics (SATPP): QCD, Effective Field Theories, and Nuclear Physics, ICTS-TIFR, 2026.
- Indian Strings Meeting, NISER Bhubaneswar and IIT Bhubaneswar, India, 2025. **Poster Presentation.**
- Loop-the-Loop-2: Feynman calculus and its applications to gravity and particle physics, online, 2025.
- 28th CAPRA Meeting on Radiation Reaction in General Relativity, University of Southampton, Southampton, United Kingdom, 2025. **Flash Talk.**
- From Asymptotic Symmetries to Flat Holography: Theoretical Aspects and Observable Consequences, INFN GGI, Florence, Italy, 2025.
- A Hundred Years of Quantum Mechanics, ICTS-TIFR, Bengaluru, India, 2025.
- Strings 2025, Annual Meeting, online.
- Hearing beyond the standard model with cosmic sources of Gravitational Waves, ICTS-TIFR, Bengaluru, India, 2024.
- Quantum Information, Quantum Field Theory, and Gravity, ICTS-TIFR, Bengaluru, India, 2024.
- Field Theory and Turbulence, ICTS-TIFR, Bengaluru, India, 2023.
- Indian Strings Meeting, IIT Bombay, India, 2023.
- Non-perturbative and Numerical Approaches to Quantum Gravity, String Theory and Holography, ICTS-TIFR, Bengaluru, India, 2022.
- IFCAM Summer School on Mathematical Aspects of Quantum Mechanics, IFCAM, IISc, Bengaluru, India, 2022.
- Physics of the Early Universe, ICTS-TIFR, Bengaluru, India, 2021.

Scholarships and Awards

- ICTS Graduate Research Fellowship. 2021
- CSIR Junior Research Fellowship. 2021
- Merit Certificate (for National Top 1 percent) in the National Graduate Physics Examination. 2016
- CBSE Central Sector Scheme of Scholarship for College and University Students. 2014